

Multiple Choice (10 marks)

1. Statement 1 | The external direct product of cyclic groups is cyclic. Statement 2 | The external direct product of D_3 and D_4 is isomorphic to D_{12} .
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1 | If R is an integral domain, then $R[x]$ is an integral domain. Statement 2 | If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)g(x)) = \deg f(x) + \deg g(x)$.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Let $(\mathbb{Z}_{10}, +, \cdot)$ be the ring of integers modulo 10, and let S be the subset of $\mathbb{Z}_{10}$ represented by $\{0, 2, 4, 6, 8\}$. Which of the following statements is FALSE?
 - (a) $(S, +, \cdot)$ is closed under addition modulo 10.
 - (b) $(S, +, \cdot)$ is closed under multiplication modulo 10.
 - (c) $(S, +, \cdot)$ has an identity under addition modulo 10.
 - (d) $(S, +, \cdot)$ has no identity under multiplication modulo 10.
4. If the finite group G contains a subgroup of order seven but no element (other than the identity) is its own inverse, then the order of G could be
 - (a) 27
 - (b) 28
 - (c) 35
 - (d) 37
5. Find the maximum possible order for some element of $\mathbb{Z}_4 \times \mathbb{Z}_6$.
 - (a) 4
 - (b) 6
 - (c) 12
 - (d) 24

True / False (10 marks)

Answer True or False

6. True or False: Every finitely generated torsion-free abelian group is not necessarily a free abelian group.
7. True or False: If a is in a subset H of a group G , then the element $a * b$ is in H for all b in G .
8. True or False: The relation $\sim$ is symmetric, meaning that if (a, b) is in $\sim$, then (b, a) must also be in $\sim$.
9. True or False: If f is a linear transformation with $f(1, 1) = 1$ and $f(-1, 0) = 2$, then $f(3, 5)$ equals 8.
10. True or False: If C is false, then it is possible for both A and B to be true.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the point of intersection of the line

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$$

and $x - y + z = 5$.

12. There are two distinct solutions x to the equation $18 + 5x^2 = 20x$. If each solution is rounded to the nearest integer, and then these two integers are multiplied together, what is the result?

End of Paper